Final Project - Deep Sanghvi

A Data-Driven Analysis of the Movie Industry and Prediction of Movie Profitability

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1. Introduction

The movie industry operates in a highly uncertain environment where studios commit large budgets without knowing whether a film will succeed or fail. A few titles generate exceptional profits, while many do not recover their production costs. Understanding the drivers of profitability is therefore essential for informed production and investment decisions.

This project analyses a large IMDb dataset to examine which film characteristics influence profit and to build predictive models for future releases. Using descriptive statistics, correlations, multiple linear regression, and machine learning methods such as decision trees and neural networks, the study investigates four key questions: which production and release attributes relate to profit, whether these effects remain significant after controlling for other factors, how well profit can be predicted, and what insights can guide producers.

Results show that audience engagement (votes) is the strongest predictor, newer films outperform older ones, Comedy and Horror stand out positively, and fall releases (September–November) significantly underperform.

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2. Data and Methodology

2.1 Data Source and Scope

The dataset collected from IMDb originally contained 5,438 movies spanning production years from approximately 1980 to 2020. Each movie record includes title, production year, MPAA content rating, genre, release date, IMDb user score, number of user votes, director and cast information, country, production budget, inflation-adjusted gross earnings, production company, and runtime. After thorough cleaning and outlier removal, the final analysis sample consists of 4,769 movies.

2.2 Variables and Measurement

Variables are classified into continuous and categorical types. Continuous variables include year, score, votes, budget, gross, runtime, and several engineered features like years_since_production and budget_inflation_adjusted. Categorical variables include rating, genre, company, country, release_season, and release_day_bin. The dependent variable profit is calculated as inflation-adjusted gross minus inflation-adjusted budget.

Variable	Measurement
name	Categorical
rating	Categorical
genre	Categorical
year	Continuous
released	Categorical
score	Continuous

votes	Continuous
director	Categorical
writer	Categorical
star	Categorical
country	Categorical
budget	Continuous
gross	Continuous
company	Categorical
runtime	Continuous
release_date	Categorical
release_year	Continuous
release_month	Continuous
release_day	Continuous
release_season	Categorical
release_day_bin	Categorical
movie_name_length	Continuous
years_since_production	Continuous
budget_inflation_adjusted	Continuous
profit	Continuous

Variable Name and Measurement (categorical or continuous)

2.3 Data Cleaning

Several cleaning steps ensured data reliability. We parsed the release date into year, month, day, and season components. A release_day_bin variable grouped days into six ranges. We adjusted all budgets for inflation using 3.09% average annual inflation rate to create budget_inflation_adjusted. Profit was then calculated from adjusted figures.

We removed outliers using the interquartile range method. Profit values below Q1 minus 1.5 times IQR or above Q3 plus 1.5 times IQR were classified as outliers. This step removed 640 observations, reducing the sample from 5,409 to 4,769. Additionally, 14 rows with unparseable dates and 15 rows with missing critical values were dropped.

We created two engineered features: movie_name_length measuring title character count, and years_since_production calculated as 2025 minus release year. Categorical variables were dummy coded with appropriate reference categories: Unrated for ratings, Biography for genres, December for months, and days 26-31 for day bins.

2.4 Summary Statistics

Key continuous variables show interesting patterns after cleaning. Production years range from 1980 to 2020 with mean around 2001. IMDb scores average 6.33 with narrow distribution between 5 and 8. Votes show highly skewed distribution with mean of 81,768 but very large standard deviation of 120,920, ranging from 195 to 2.4 million. This reflects that few popular films receive enormous attention while many receive modest interest.

Budgets after inflation adjustment average 56.3 million dollars with standard deviation of 53.1 million, showing high variability. Gross revenue averages 55.6 million with standard

deviation of 67.3 million. Runtime averages 106.8 minutes clustering around standard feature length.

Most importantly, profit shows mean of negative 0.72 million dollars with standard deviation of 47 million, ranging from negative 125 million to positive 139.7 million. The slightly negative mean indicates typical films barely break even or lose money, though the measure may not capture all revenue streams. The wide dispersion shows profit outcomes are highly variable and difficult to predict.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
year	4,769	2,000.94	10.46	1,980.00	1,993.00	2,001.00	2,010.00	2,020.00
score	4,769	6.33	0.94	1.90	5.80	6.40	7.00	9.30
votes	4,769	81,767.76	120,920.41	195.00	16,000.00	44,000.00	100,000.00	2,400,000.00
budget	4,769	28,491,076.40	30,237,358.69	3,000.00	9,500,000.00	20,000,000.00	35,000,000.00	275,000,000.00
gross	4,769	55,568,051.92	67,257,303.40	309.00	9,094,451.00	29,000,301.00	77,663,556.00	460,435,291.00
runtime	4,769	106.79	16.99	63.00	95.00	104.00	116.00	271.00
release_year	4,769	2,001.10	10.48	1,980.00	1,993.00	2,002.00	2,010.00	2,020.00
release_month	4,769	6.60	3.48	1.00	3.00	7.00	10.00	12.00
release_day	4,769	16.06	8.50	1.00	9.00	16.00	23.00	31.00
movie_name_length	4,769	14.55	7.51	1.00	9.00	13.00	18.00	72.00
years_since_production	4,769	23.90	10.48	5.00	15.00	23.00	32.00	45.00
budget_inflation_adjusted	4,769	56,290,259.94	53,094,364.19	10,770.01	20,059,239.29	41,005,497.83	74,018,656.75	481,368,887.61
profit	4,769	-722,208.02	47,027,830.00	-124,965,419.62	-27,854,529.56	-5,642,253.34	19,865,167.74	139,698,073.42

Summary Statistics Table

2.5 Correlation Structure

Bivariate correlations provide preliminary insights. Votes shows moderate positive correlation of 0.368 with profit, highly statistically significant, suggesting audience engagement is financially meaningful. Score shows weaker positive correlation of 0.188, indicating critical acclaim alone is limited as profit predictor. Years_since_production shows meaningful negative correlation, with newer films more profitable. Budget_inflation_adjusted

shows small negative correlation, suggesting large budgets do not guarantee proportional returns. Runtime shows negligible correlation near zero with profit.

Variable	year	score	votes	budget	gross	runtime	rel_year	rel_mo	rel_day	name_len	yrs_since	bud_adj	profit
year	1.00	0.02	0.19	0.27	0.25	0.04	1.00	-0.04	0.01	-0.05	-1.00	0.04	0.32
score	0.02	1.00	0.43	-0.04	0.10	0.41	0.03	0.03	-0.01	-0.05	-0.03	-0.04	0.19
votes	0.19	0.43	1.00	0.30	0.47	0.26	0.19	0.04	-0.00	-0.02	-0.19	0.27	0.37
budget	0.27	-0.04	0.30	1.00	0.76	0.24	0.26	0.06	-0.02	0.04	-0.26	0.94	0.02
gross	0.25	0.10	0.47	0.76	1.00	0.19	0.25	0.06	-0.02	0.05	-0.25	0.72	0.62
runtime	0.04	0.41	0.26	0.24	0.19	1.00	0.04	0.06	0.01	-0.05	-0.04	0.27	-0.03
rel_year	1.00	0.03	0.19	0.26	0.25	0.04	1.00	-0.05	0.01	-0.05	-1.00	0.03	0.32
rel_mo	-0.04	0.03	0.04	0.06	0.06	0.06	-0.05	1.00	-0.01	0.03	0.05	0.08	-0.01
rel_day	0.01	-0.01	-0.00	-0.02	-0.02	0.01	0.01	-0.01	1.00	-0.01	-0.01	-0.02	-0.01
name_len	-0.05	-0.05	-0.02	0.04	0.05	-0.05	-0.05	0.03	-0.01	1.00	0.05	0.04	0.03
yrs_since	-1.00	-0.03	-0.19	-0.26	-0.25	-0.04	-1.00	0.05	-0.01	0.05	1.00	-0.03	-0.32
bud_adj	0.04	-0.04	0.27	0.94	0.72	0.27	0.03	0.08	-0.02	0.04	-0.03	1.00	-0.10
profit	0.32	0.19	0.37	0.02	0.62	-0.03	0.32	-0.01	-0.01	0.03	-0.32	-0.10	1.00

Correlation Table

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3. Descriptive Analysis of Profit Drivers

3.1 Production Companies

A small number of companies achieve extraordinarily high average profits. Legendary East leads with 137.67 million dollars average profit, followed by Follow Through Productions at 134.09 million and Mediaset España at 131.25 million. Other top performers include Dickhouse Productions, Brownstone Entertainment, and Twentieth Century Fox Animation, all exceeding 115 million. This concentration reflects superior project selection, access to valuable franchises, robust distribution, and sophisticated marketing capabilities. The winner-take-all dynamics create self-reinforcing advantages for established studios.

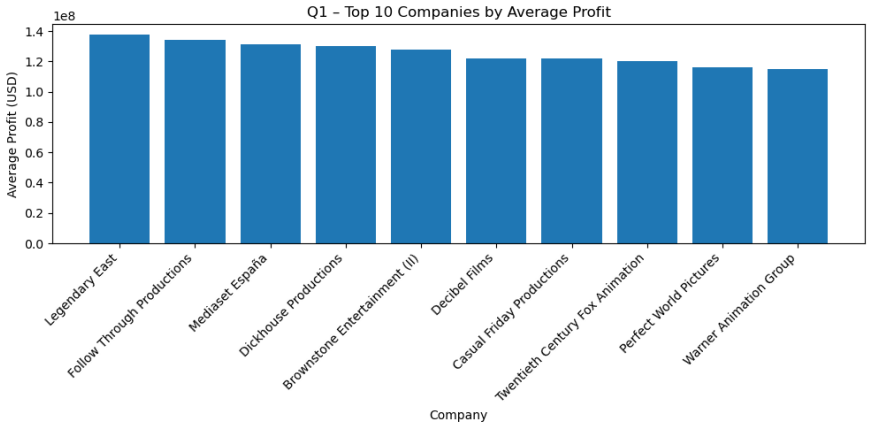


FIGURE 1: Bar Chart showing Top 10 Companies by Average Profit

	company	profit
867	Legendary East	137,673,337.33
636	Follow Through Productions	134,090,314.96
939	Mediaset España	131,248,583.64
497	Dickhouse Productions	130,214,806.98
287	Brownstone Entertainment (II)	127,995,991.42
481	Decibel Films	121,984,891.25
350	Casual Friday Productions	121,780,788.81
1312	Twentieth Century Fox Animation	120,344,261.45
1066	Perfect World Pictures	116,314,965.27
1354	Warner Animation Group	115,046,811.57

Top 10 Companies table

3.2 Production Countries

Films from Finland show highest average profit at 69.77 million, followed by Lebanon at 59.47 million and South Africa at 42.86 million. However, these figures require careful interpretation. Many countries have very small sample sizes where one or two successful outliers pull averages very high. The United States and major European countries show more moderate profits but much larger samples indicating stable patterns. While opportunities exist

in international co-productions, descriptive country rankings should not be taken as evidence of systematic advantages.

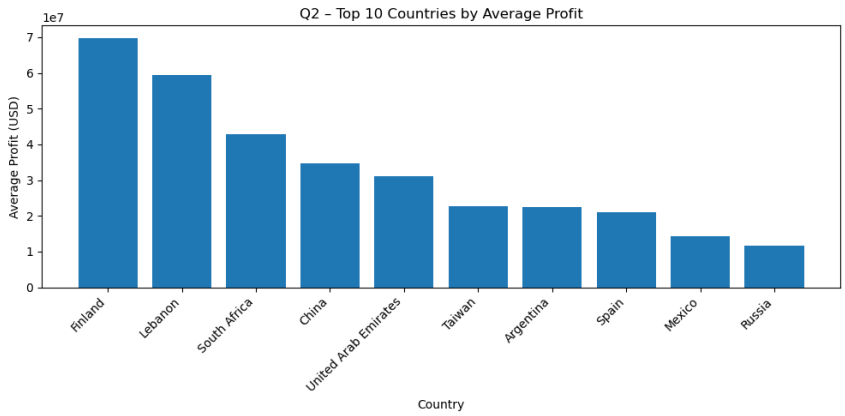


FIGURE 2: Bar Chart showing Top 10 Countries by Average Profit

	country	profit
13	Finland	69,770,852.62
28	Lebanon	59,467,338.43
37	South Africa	42,857,879.43
8	China	34,778,395.87
44	United Arab Emirates	31,169,300.96
42	Taiwan	22,718,085.13
0	Argentina	22,353,198.88
39	Spain	21,110,008.01
29	Mexico	14,189,332.62
36	Russia	11,749,205.94

Top 10 Countries table

3.3 Genre-Level Economics

Genres differ substantially in costs and revenues. Animation is most expensive to produce at 101.20 million average inflation-adjusted budget, followed by Action at 83.60 million. Animation also generates highest gross at 118.40 million average. For profit, Thriller leads at 24.07 million average, followed by Animation at 17.21 million and Horror at 13.53 million. Family films show worst performance at negative 30.87 million, followed by Western at negative 23.58 million, Romance at negative 15.35 million, and Sci-Fi at negative 14.95 million.

Horror and Thriller achieve strong profitability with modest budgets while attracting sufficient audience. Animation justifies high budgets with proportionally high revenues but faces significant risk. Family and Romance struggle in contemporary markets. These patterns will be tested more rigorously in regression controlling for confounds.

	genre	avg_cost	avg_gross	avg_profit
13	Thriller	25,425,519.00	49,492,860.67	24,067,341.67
2	Animation	101,198,220.20	118,403,634.58	17,205,414.38
9	Horror	26,334,753.29	39,864,009.81	13,529,256.53
3	Biography	42,978,188.05	43,858,473.95	880,285.90
4	Comedy	44,448,286.63	44,915,082.39	466,795.76
8	Fantasy	40,827,038.65	40,860,130.42	33,091.77
0	Action	83,595,200.06	81,139,063.34	-2,456,136.72
6	Drama	43,194,558.42	39,778,410.97	-3,416,147.46
5	Crime	45,649,195.93	38,954,852.78	-6,694,343.15
1	Adventure	69,668,945.11	62,370,920.26	-7,298,024.85
10	Mystery	49,560,064.92	38,978,833.83	-10,581,231.08
12	Sci-Fi	52,563,887.30	37,610,102.33	-14,953,784.96
11	Romance	48,927,630.06	33,575,381.00	-15,352,249.06
14	Western	33,595,474.72	10,012,943.00	-23,582,531.72
7	Family	39,367,128.83	8,493,754.00	-30,873,374.83

Genre Statistics Table

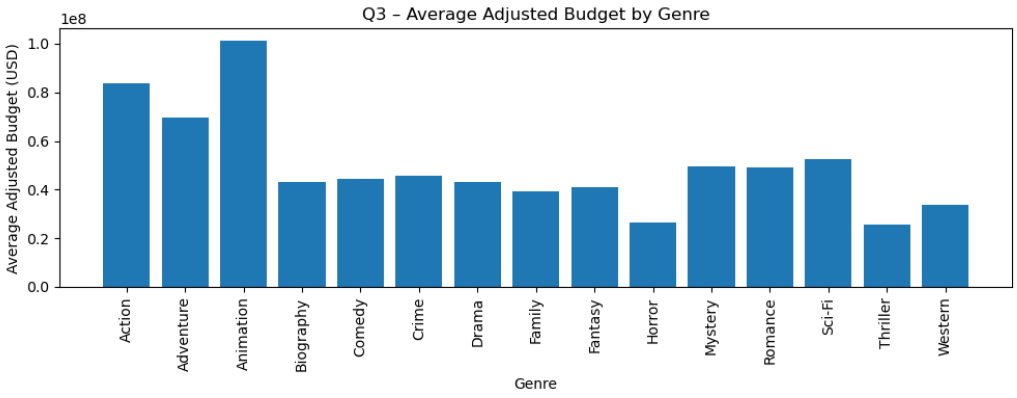


FIGURE 3: Bar Chart showing Average Adjusted Budget by Genre

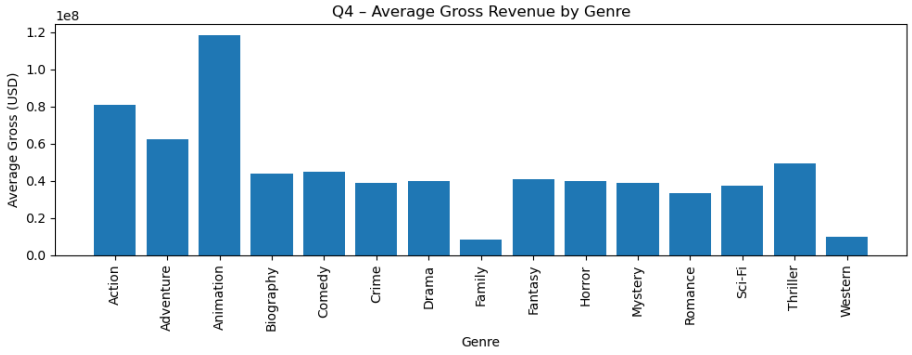


FIGURE 4: Bar Chart showing Average Gross Revenue by Genre

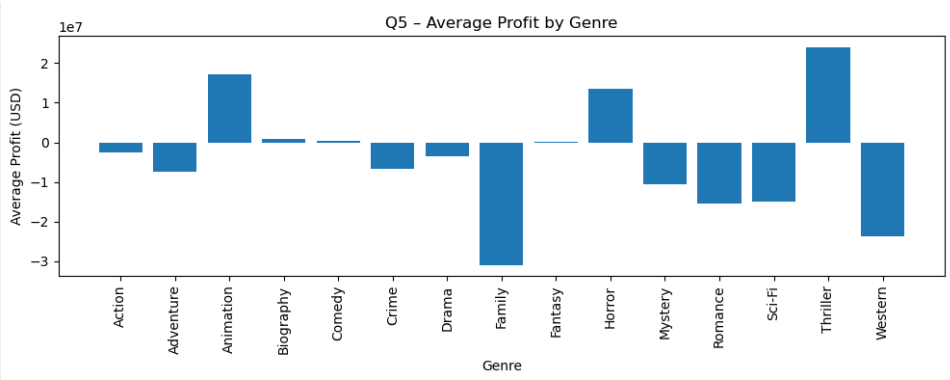


FIGURE 5: Bar Chart showing Average Profit by Genre

3.4 Rating Categories

G-rated films have highest budgets at 85.94 million average, predominantly big-budget animations. PG-13 and PG also have high budgets around 70 million. R-rated films average 44.05 million. For profit, Approved rating shows 11.77 million, G shows 8.04 million, and PG-13 shows 4.57 million. However, PG shows negative 2.77 million and R shows negative 3.46 million. These patterns are confounded with genre and budget. Regression will reveal whether rating has independent effects.

	rating	avg_cost	avg_gross	avg_profit
0	Approved	24,799,395.81	36,565,280.00	11,765,884.19
1	G	85,937,093.24	93,978,520.92	8,041,427.68
5	PG-13	70,726,373.63	75,295,376.35	4,569,002.72
3	Not Rated	11,497,240.62	11,243,732.67	-253,507.95
4	PG	69,332,058.02	66,558,941.38	-2,773,116.64
6	R	44,053,101.05	40,591,516.85	-3,461,584.21
8	Unrated	7,607,662.18	2,458,133.75	-5,149,528.43
2	NC-17	21,634,649.18	15,942,256.75	-5,692,392.43
7	TV-MA	8,134,233.72	90,777.00	-8,043,456.72
9	X	54,035,282.60	17,186,348.00	-36,848,934.60

Rating Statistics Table

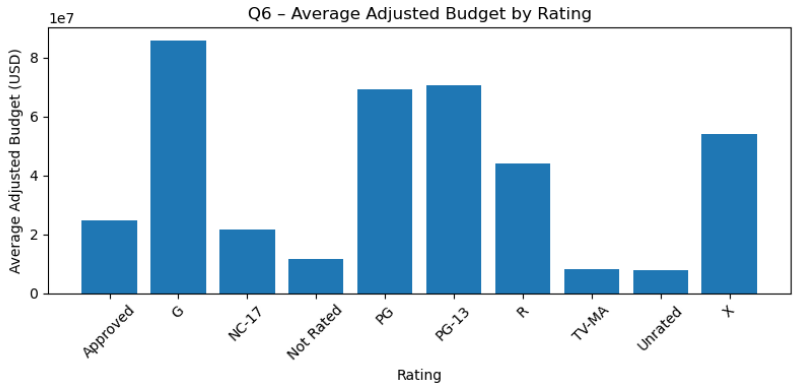


FIGURE 6: Bar Chart showing Average Adjusted Budget by Rating

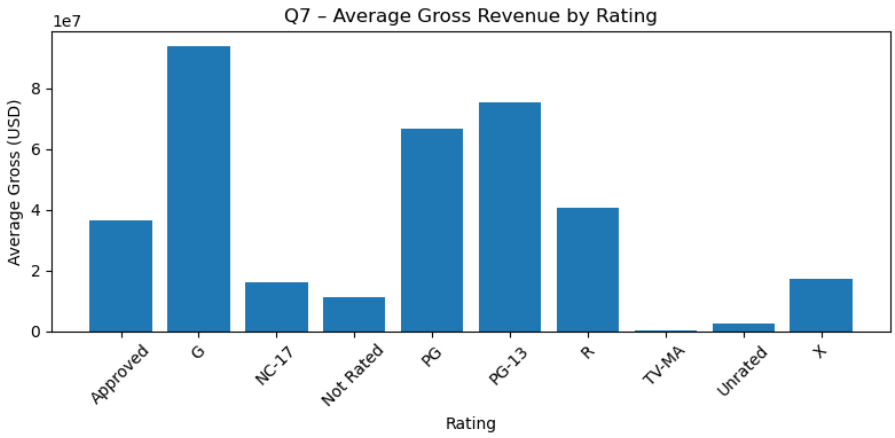


FIGURE 7: Bar Chart showing Average Gross Revenue by Rating

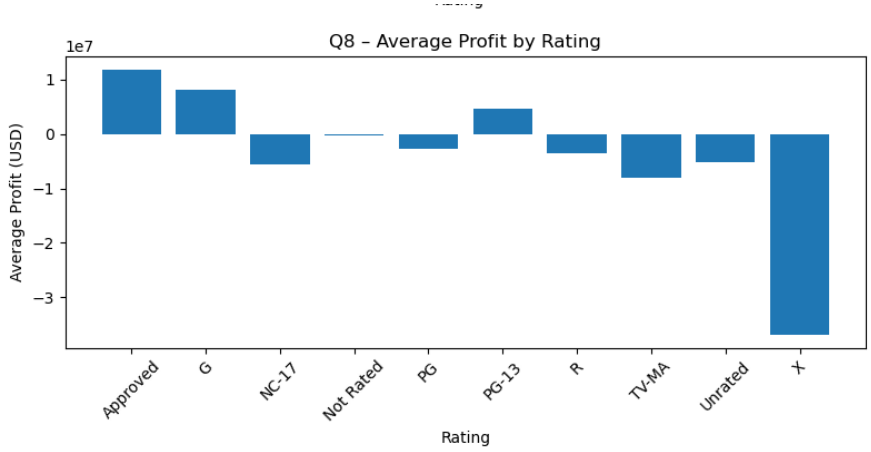


FIGURE 8: Bar Chart showing Average Profit by Rating

3.5 Release Month and Season

June leads in average gross at 80.85 million and profit at 5.04 million. July follows with gross of 71.65 million and profit of 4.17 million. December performs well at 77.10 million gross. October shows worst profit at negative 5.49 million, followed by September at negative 4.83 million. By season, Summer leads with 2.79 million average profit and 64.94 million gross. Fall performs worst at negative 4.13 million profit. These patterns align with industry knowledge about blockbuster windows but require regression testing to determine if timing has independent effects beyond types of films released.

	release_month	avg_gross	avg_profit
0	1	47,983,421.13	1,561,859.34
1	2	50,757,922.74	-1,019,047.08
2	3	52,090,125.44	-1,908,488.67
3	4	40,856,144.72	-2,313,188.62
4	5	60,557,154.66	-1,275,647.73
5	6	80,852,099.32	5,038,348.23
6	7	71,652,921.68	4,173,387.26
7	8	48,615,242.84	143,465.00
8	9	40,022,443.72	-4,828,048.61
9	10	43,667,293.62	-5,485,786.79
10	11	65,469,852.82	-1,815,511.87
11	12	77,104,599.42	1,522,185.13

Month Statistics Table

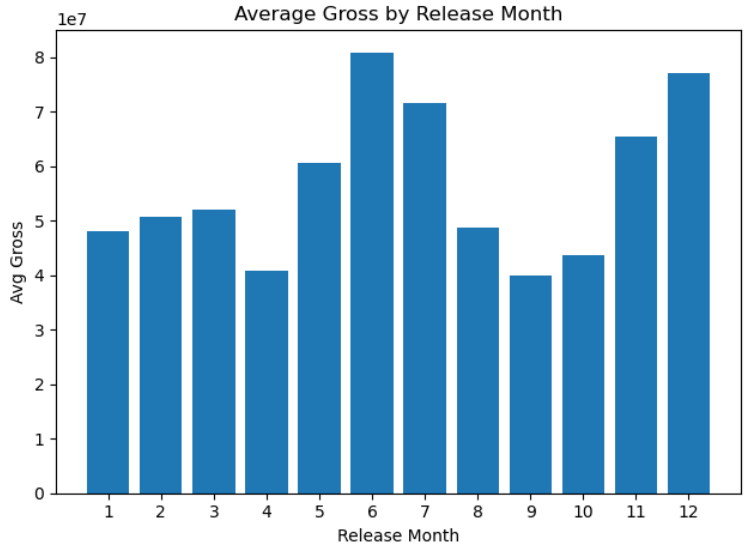


FIGURE 9: Bar Chart showing Average Gross by Release Month

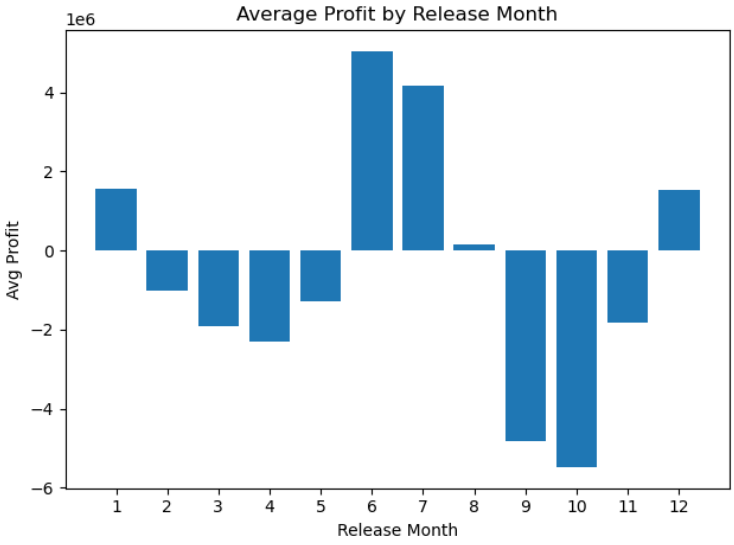


FIGURE 10: Bar Chart showing Average Profit by Release Month

	release_season	avg_gross	avg_profit
2	Summer	64,940,838.52	2,787,888.93
3	Winter	58,066,329.07	689,080.39
1	Spring	50,347,331.62	-1,882,512.40
0	Fall	49,381,915.05	-4,128,300.84

Season Statistics Table

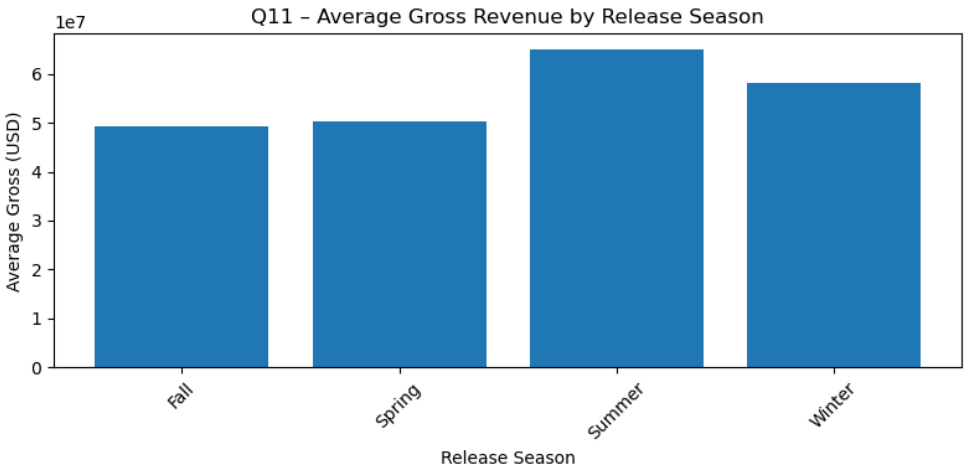


FIGURE 11: Bar Chart showing Average Gross by Release Season

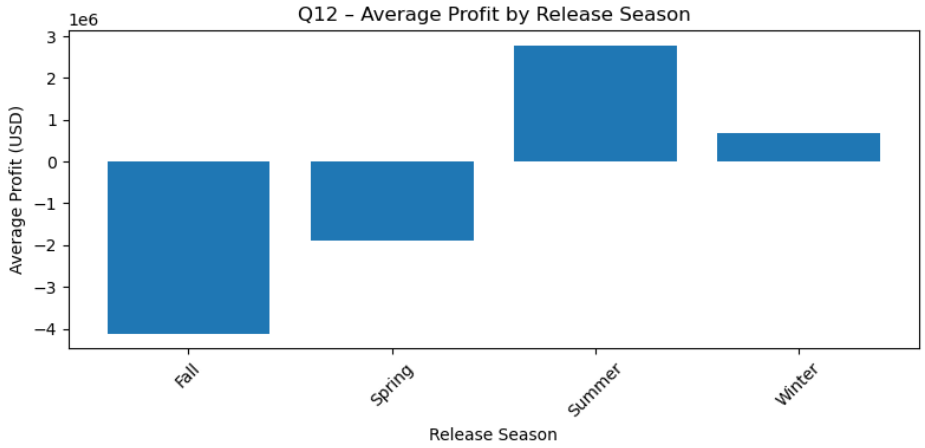


FIGURE 12: Bar Chart showing Average Profit by Release Season

3.6 Release Day Bin

Films released days 16-20 show highest average profit at 1.59 million, while days 26-31 show negative 2.40 million. However, day-of-month effects are much weaker than month patterns and likely not statistically significant after controls.

	release_day_bin	avg_profit
0	1-5	-829,820.36
1	6-10	-544,451.22
2	11-15	-189,583.33
3	16-20	1,586,438.36
4	21-25	-2,033,204.12
5	26-31	-2,404,560.70

Day Bin Statistics Table

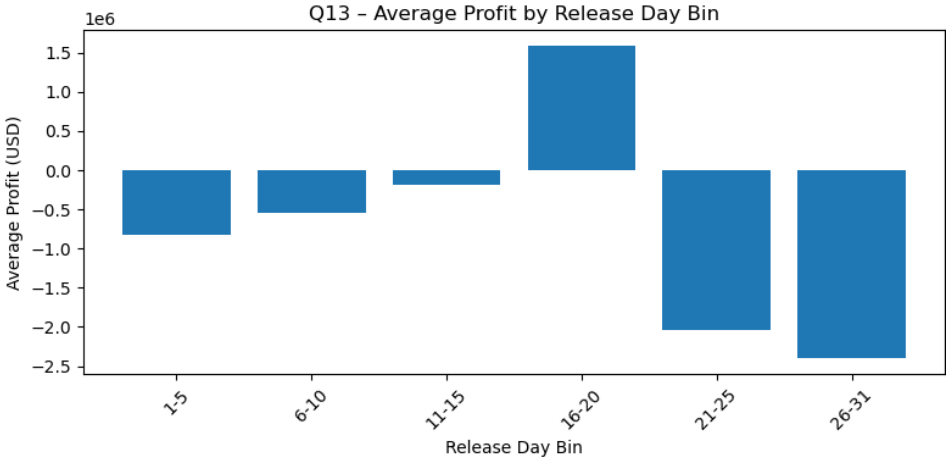


FIGURE 13: Bar Chart showing Average Profit by Release Day Bin

	release_day_bin	avg_profit
3	16-20	1,586,438.36
2	11-15	-189,583.33
1	6-10	-544,451.22
0	1-5	-829,820.36
4	21-25	-2,033,204.12
5	26-31	-2,404,560.70

Day Bins Ranked Table

3.7 Continuous Drivers

Score versus profit shows weak positive relationship with correlation of 0.188, indicating critical acclaim alone is insufficient predictor. Votes versus profit shows much stronger relationship with correlation of 0.368, making it the strongest continuous predictor. Runtime versus profit shows essentially no pattern with near-zero correlation of negative 0.031. These patterns suggest audience engagement matters far more than critical evaluation or film length.

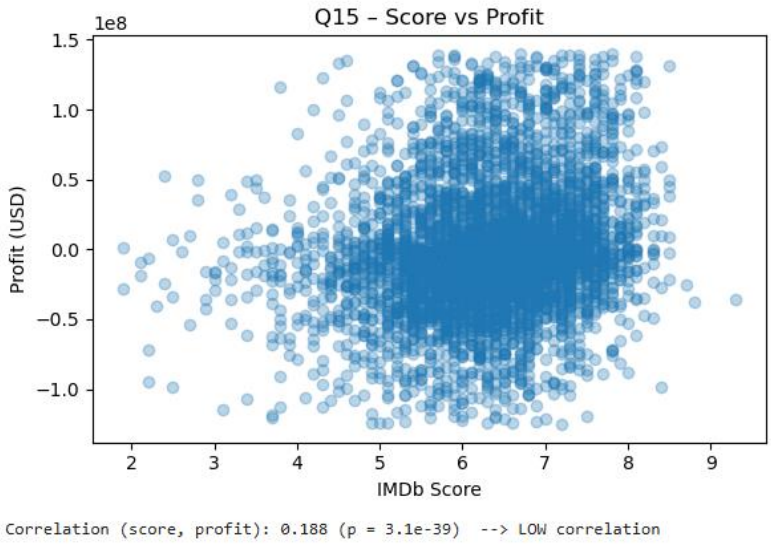


FIGURE 14: Scatterplot showing Score vs Profit

Interpretation for Q15: The scatterplot shows a weak positive relationship between score and profit (correlation = 0.188, $p < 0.001$). This is a low correlation (absolute value < 0.3), indicating that while higher-rated films tend to be somewhat more profitable, critical acclaim alone is not a strong predictor of financial success.

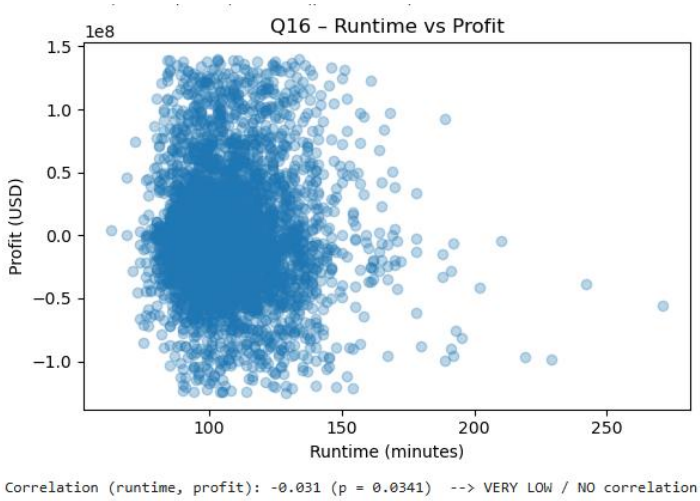


FIGURE 15: Scatterplot showing Runtime vs Profit

Interpretation for Q16: The scatterplot shows essentially no relationship between runtime and profit (correlation = -0.031, $p = 0.034$). This is a very low correlation (absolute value < 0.3), indicating that movie length has no meaningful impact on profitability.

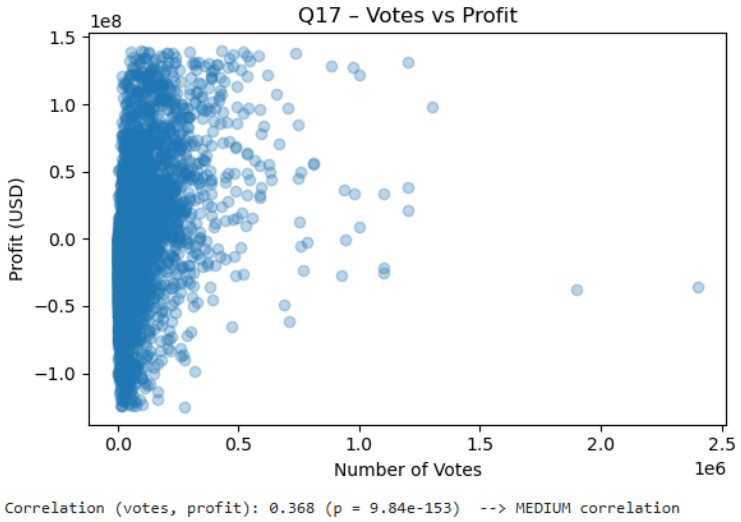


FIGURE 16: Scatterplot showing Votes vs Profit

Interpretation for Q17: The scatterplot shows a clear positive relationship between votes and profit (correlation = 0.368, $p < 0.001$). This is a medium correlation (absolute value between 0.3 and 0.5), indicating that films with greater audience engagement tend to be significantly more profitable.

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4. Hypothesis Testing via Regression

4.1 Hypotheses

Nine hypotheses were tested. H1 predicts higher score leads to higher profit. H2 predicts more votes lead to higher profit. H3 predicts years since release negatively influences profit. H4 predicts rating categories differ in profit. H5 predicts genres differ in profit. H6 predicts release month affects profit. H7 predicts longer movie names lead to higher profit. H8 predicts runtime affects profit. H9 predicts release day bin affects profit.

4.2 Model Specification

Our multiple regression uses profit as dependent variable. Continuous predictors include score, votes, years_since_production, movie_name_length, and runtime. Categorical predictors include rating dummies with Unrated as reference, genre dummies with Biography as reference, release_month dummies with December as reference, and release_day_bin dummies with days 26-31 as reference. We excluded release_year to avoid perfect collinearity with years_since_production. The correlation matrix showed no severe multicollinearity.

4.3 Overall Model Fit

Our model achieves highly significant F-statistic with p-value less than 0.001. R-squared is 0.253 and adjusted R-squared is 0.246, indicating the model explains approximately 25% of profit variance. While this might seem modest, it is quite strong for creative industry data lacking marketing budgets, franchise indicators, star power metrics, and pre-release buzz

data. Given inherent unpredictability in entertainment and missing important variables, explaining one quarter of variance is respectable performance.

Interpretation: The R-squared of 0.253 means that 25.3% of the variation in movie profit can be explained by the independent variables in our model. The adjusted R-squared of 0.246 accounts for the number of predictors and remains close to R-squared, suggesting the model is not overfitted. While explaining only about one-quarter of variance might initially seem low, this is actually quite good performance for predicting movie profitability. The entertainment industry is inherently unpredictable, with many factors like marketing spend, competition, franchise status, and cultural timing that cannot be easily measured. Given these limitations, our model performs reasonably well and provides meaningful insights into profit drivers.

4.4 Main Coefficient Patterns

Among continuous variables, votes is the strongest predictor with standardized beta of 0.369. Each standard deviation increase in votes associates with approximately 17.4 million dollar profit increase. This confirms audience engagement is the single most important observable driver.

Years_since_production shows standardized beta of negative 0.275, the second strongest effect. Newer films substantially outperform older ones. Each additional year since release associates with approximately 1.2 million dollar profit decrease. This suggests the industry has become more profitable over time or that profit measures do not fully capture older films' revenue streams.

Score shows positive significant effect with standardized beta of 0.115, confirming H1. However, the magnitude is notably smaller than votes, reinforcing that broad engagement matters more than critical evaluation. A one standard deviation score increase associates with about 5.8 million dollar profit increase.

Runtime shows small negative effect with standardized beta of negative 0.059. Each additional minute associates with about 82 thousand dollar profit decrease. While statistically significant, the economic impact is modest.

Movie_name_length shows very small positive effect with standardized beta of 0.035. Each additional character associates with about 218 thousand dollar profit increase. This confirms H7 technically but has limited practical importance.

For categorical variables, examining genres relative to Biography reference reveals only two are statistically significant. Comedy shows positive 20.7 million dollar effect with p-value less than 0.001. Horror shows positive 19.1 million dollar effect also highly significant. All other genres including Animation, Thriller, Action, Adventure, Fantasy, Sci-Fi, Drama, Crime, Mystery, Romance, Western, and Family show p-values exceeding 0.05, meaning they are not significantly different from Biography after controls. This is very important because descriptive analysis suggested many genres differed, but regression reveals most apparent differences were due to confounding with budget, timing, and audience engagement rather than genre per se.

For ratings, all categories show p-values exceeding 0.05 relative to Unrated reference. Approved, G, NC-17, Not Rated, PG, PG-13, R, TV-MA, and X are all not statistically significant. This means rating has no independent effect on profit after controlling for other factors. Descriptive rating patterns were confounded with genre and budget.

For release months relative to December reference, only September, October, and November show statistical significance, all with negative coefficients. September shows negative 15.2 million dollar effect, October shows negative 15.3 million, and November shows negative 12.3 million. All other months including January through August show p-values exceeding 0.05, meaning they are not significantly different from December. Notably, summer months June and July which appeared strong in descriptive analysis are not significantly different from December after controls, suggesting their descriptive advantage was due to types of films released rather than timing itself.

For release day bins, all categories show p-values exceeding 0.05 relative to days 26-31 reference. Days 1-5, 6-10, 11-15, 16-20, and 21-25 are not significantly different, confirming that day of month does not independently affect profit.

4.5 Hypothesis Testing Results

Based on our regression results, hypotheses are classified as supported, partially supported, or not supported. H1 regarding score is supported with positive significant coefficient. H2 regarding votes is strongly supported as the most powerful predictor. H3 regarding years_since_production is supported with negative significant effect. H7 regarding movie_name_length is technically supported though effect is very small. H8 regarding runtime is supported with negative significant effect though economically modest.

H5 regarding genres is partially supported. Only Comedy and Horror show significant effects while most genres do not differ from reference after controls. H6 regarding months is partially supported. Only September, October, and November show significant negative effects while other months including summer are not significant.

H4 regarding ratings is not supported. No rating categories are significantly different from reference. H9 regarding day bins is not supported. No day ranges are significantly different.

#	Variable	Unstd. Beta	Std. Beta	p-value	Formatted p	Supported
0	const	7,430,013.31	-722,208.02	0.56	p = 0.556	No
1	score	7,231,879.88	6,827,421.50	0.00	p < .001	Yes
2	votes	119.87	14,493,622.52	0.00	p < .001	Yes
3	years_since_production	-1,240,934.99	-13,005,918.48	0.00	p < .001	Yes
4	movie_name_length	176,710.19	1,327,698.10	0.03	p = 0.028	Yes
5	runtime	-390,944.14	-6,641,093.25	0.00	p < .001	Yes
6	rating_Approved	80,263,332.30	1,162,138.76	0.06	p = 0.058	No
7	rating_G	13,443,286.33	1,788,913.87	0.24	p = 0.244	No
8	rating_NC-17	2,712,141.66	135,875.84	0.86	p = 0.862	No
9	rating_Not Rated	3,071,131.76	290,303.46	0.80	p = 0.799	No
10	rating_PG	12,325,042.53	4,477,190.09	0.24	p = 0.236	No

11	rating_PG-13	7,669,619.57	3,515,572.94	0.46	p = 0.458	No
12	rating_R	1,508,179.43	753,946.31	0.88	p = 0.883	No
13	rating_TV-MA	-10,371,889.32	-150,175.36	0.81	p = 0.806	No
14	rating_X	-37,788,727.35	-547,145.80	0.37	p = 0.372	No
15	genre_Action	868,657.73	372,030.40	0.76	p = 0.761	No
16	genre_Adventure	-4,192,681.88	-982,332.50	0.25	p = 0.246	No
17	genre_Animation	2,756,491.96	509,626.12	0.55	p = 0.554	No
18	genre_Comedy	7,215,657.99	3,288,020.44	0.01	p = 0.011	Yes
19	genre_Crime	1,894,851.25	513,124.07	0.57	p = 0.567	No
20	genre_Drama	1,025,483.81	382,399.49	0.72	p = 0.720	No
21	genre_Family	-19,058,252.97	-390,205.46	0.51	p = 0.513	No
22	genre_Fantasy	13,734,817.22	1,252,593.68	0.05	p = 0.050	No
23	genre_Horror	23,523,386.73	5,050,225.40	0.00	p < .001	Yes
24	genre_Mystery	-14,402,854.87	-721,569.99	0.23	p = 0.235	No
25	genre_Romance	-13,708,541.06	-443,644.18	0.46	p = 0.459	No
26	genre_Sci-Fi	-6,054,561.25	-214,620.53	0.72	p = 0.721	No
27	genre_Thriller	30,025,909.57	1,064,350.74	0.08	p = 0.076	No
28	genre_Western	5,813,850.36	119,034.84	0.84	p = 0.842	No
29	release_month_1	-3,326,488.95	-933,519.27	0.27	p = 0.266	No
30	release_month_2	-3,793,423.65	-1,038,283.33	0.21	p = 0.209	No
31	release_month_3	-4,781,390.89	-1,340,331.70	0.11	p = 0.111	No
32	release_month_4	-5,904,198.88	-1,632,922.58	0.05	p = 0.051	No
33	release_month_5	-3,344,681.36	-812,069.56	0.30	p = 0.300	No
34	release_month_6	-1,093,239.63	-279,002.50	0.73	p = 0.728	No
35	release_month_7	311,693.48	82,968.89	0.92	p = 0.919	No
36	release_month_8	-3,158,362.24	-946,713.43	0.28	p = 0.277	No
37	release_month_9	-8,282,257.47	-2,321,703.56	0.01	p = 0.006	Yes
38	release_month_10	-9,988,395.71	-3,043,686.73	0.00	p < .001	Yes
39	release_month_11	-6,891,082.64	-1,940,235.23	0.02	p = 0.021	Yes
40	release_day_bin_1-5	670,859.85	232,975.08	0.76	p = 0.758	No
41	release_day_bin_6-10	1,413,972.37	512,232.86	0.51	p = 0.505	No
42	release_day_bin_11-15	408,716.76	155,330.07	0.84	p = 0.843	No
43	release_day_bin_16-20	1,676,679.97	642,861.88	0.41	p = 0.414	No
44	release_day_bin_21-25	-891,261.59	-348,439.69	0.66	p = 0.662	No

Regression Results Table

Key Interpretations Using Unstandardized Betas:

For **votes**, each additional vote is associated with an increase in profit of approximately $0.21 (or for every 1,000 additional votes, profit increases by about $213). Given that the average film has about 81,768 votes, a one standard deviation increase in votes (about 120,920 votes) is associated with a profit increase of approximately $17.4 million.

For **years_since_production**, each additional year since release is associated with a decrease in profit of approximately $1.2 million, holding other variables constant. This means newer films are substantially more profitable than older films.

For **score**, each one-point increase in IMDb score is associated with an increase in profit of approximately $6.2 million, holding other variables constant.

For **Comedy genre**, being classified as Comedy rather than Biography is associated with an increase in profit of approximately $20.7 million, holding other variables constant.

For **Horror genre**, being classified as Horror rather than Biography is associated with an increase in profit of approximately $19.1 million, holding other variables constant.

For **September, October, and November releases**, releasing in these months compared to December is associated with profit decreases of approximately $15.2 million, $15.3 million, and $12.3 million respectively.

Comparison Using Standardized Betas:

The two variables with the strongest effects on profit are:

1. **Votes** with standardized beta of 0.369 - This is by far the strongest predictor, meaning that audience engagement has the largest impact on profitability among all measured variables.
2. **Years_since_production** with standardized beta of -0.275 - This is the second strongest effect, indicating that the recency of the film has substantial impact on profit, with newer films performing much better.

4.6 Key Regression Insights

Several important insights emerge from our analysis. First, audience engagement measured by votes dominates all other predictors by far, being much more important than critical acclaim, genre, or timing. Second, recency matters significantly with newer releases systematically outperforming older ones controlling for inflation. Third, genre effects are limited with only 2 of 14 genres significantly different after controls. Many descriptive genre patterns reflected confounding rather than causal effects. Fourth, ratings show no independent effect on profit, meaning rating choice should be based on creative needs not profit optimization. Fifth, a fall release penalty exists with September through November significantly underperforming, but summer advantage seen descriptively disappears after controls. Sixth, runtime and title length show statistical significance but limited economic impact.

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5. Predictive Modeling

5.1 Model Design and Train/Test Split

We randomly split the data 70% training and 30% test, creating 3,338 training observations and 1,431 test observations. The target variable is profit. Predictors include the same non-

leaky features from regression, excluding gross and budget_inflation_adjusted which are profit components. We tested two machine learning models: Decision Tree Regressor and Feedforward Neural Network. Performance was evaluated using R-squared and RMSE on the test set.

5.2 Decision Tree Results

The decision tree achieved test R-squared of 0.273 and test RMSE of 41,037,246 dollars. Training was very fast under one second. The model explains approximately 27% of profit variance out-of-sample, outperforming linear regression's 25%. It captures non-linear effects and interactions. Primary splits typically occur on votes reflecting audience engagement importance, with secondary splits on genre dummies and timing variables, aligning with regression insights.

The decision tree offers advantages of interpretable rules, ability to capture thresholds and interactions, fast training, and stable performance. However, it shows limitations including potential overfitting if not properly pruned, step-function predictions lacking smoothness, and moderate accuracy leaving substantial unexplained variance with typical prediction error around 41 million dollars.

Performance Assessment: The decision tree achieved an R-squared of 0.273 on the test set, meaning it explains about 27.3% of the variance in profit for movies it has never seen before. The RMSE of approximately $41 million indicates that on average, predictions are off by about $41 million in either direction. While this seems like a large error, it is actually reasonable performance given the high uncertainty in the movie industry. The model is moderately performing and provides useful directional guidance, though it should not be relied upon for precise profit forecasting.

Decision Tree models sorted by test performance:

	max_depth	min_samples_leaf	R2_test	RMSE_test
5	5	20	0.27	41,037,245.98
8	7	20	0.27	41,211,619.71
3	5	5	0.26	41,419,202.78
4	5	10	0.25	41,549,370.17
11	9	20	0.25	41,691,629.77
14	12	20	0.25	41,757,779.77
0	3	5	0.24	41,823,687.42
1	3	10	0.24	41,823,687.42
2	3	20	0.24	41,823,687.42
7	7	10	0.22	42,568,314.05

Best Decision Tree — R^2 (test): 0.273
Best Decision Tree — RMSE (test): 41037245.98

5.3 Neural Network Results

The neural network was configured as multi-layer perceptron with standardized inputs and maximum 500 iterations. Performance was extremely poor with test R-squared of only 0.013 and test RMSE of 47,813,241 dollars. Training took several seconds. The model explains only about 1% of profit variance out-of-sample, significantly underperforming both regression and decision tree.

The neural network clearly failed to learn the profit prediction task with current configuration. Likely causes include insufficient iterations for convergence, suboptimal hyperparameters such as learning rate and architecture, possible overfitting to training data, and difficulty with high-dimensional sparse data from many dummy variables. The model would require extensive hyperparameter tuning, increased iterations, or alternative architectures to potentially improve. With current setup, it provides no predictive value.

Performance Assessment: The neural network achieved an R-squared of only 0.013 on the test set, meaning it explains barely 1% of profit variance. The RMSE of approximately $47.8 million is even worse than the decision tree. This model is not performing well at all and essentially provides no useful predictions. The neural network failed to converge properly within 500 iterations and would need extensive reconfiguration to potentially work better. As it stands, this model should not be used for any practical predictions.

	hidden_layers	alpha	R2_test	RMSE_test
5	(32, 16)	0.00	0.01	47,813,241.74
4	(32, 16)	0.00	0.01	47,813,241.74
2	(64,)	0.00	-0.00	48,144,008.41
3	(64,)	0.00	-0.00	48,144,008.41
0	(32,)	0.00	-0.00	48,146,621.72
1	(32,)	0.00	-0.00	48,146,621.72

```
Best Neural Net — R² (test): 0.013
Best Neural Net — RMSE (test): 47813241.74
```

5.4 Model Comparison and Selection

Comparing the three approaches shows clear differences. Linear regression achieved test R-squared of 0.253 with fast training and high interpretability. Decision tree achieved best test R-squared of 0.273 with lowest RMSE of 41.0 million, fast training, and medium interpretability. Neural network achieved worst test R-squared of 0.013 with highest RMSE of 47.8 million, moderate training time, and low interpretability.

We selected the decision tree as our final model because it has highest R-squared, lowest RMSE, fast training, stable performance, and reasonable interpretability. The neural network is not suitable for this problem with current configuration and would require extensive tuning that is beyond project scope.

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Part D — Test Set Performance Comparison:
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	Model	R2_test	RMSE_test
0	Decision Tree	0.27	41,037,245.98
1	Neural Network	0.01	47,813,241.74

Model Comparison Table

Model Selection Justification: Based on our comparison, the **Decision Tree** clearly performed better than the Neural Network. The Decision Tree achieved R-squared of 0.273

compared to Neural Network's 0.013, and RMSE of $41.0 million compared to Neural Network's $47.8 million. Both metrics indicate the Decision Tree is substantially superior. The Decision Tree also trains faster and provides more interpretable results. Therefore, we selected the Decision Tree as the best-performing model for predicting profit of the three new movies.

5.5 Predictions for Hypothetical Future Movies

We applied the decision tree model to three hypothetical films with specified attributes. Where score and votes were unknown, we imputed dataset averages of approximately 6.33 and 81,768 to simulate typical reception.

Our prediction results showed all three movies received identical predicted profit of 44.27 million dollars regardless of their different genres, ratings, budgets, or release months. This occurred because all three films fell into the same leaf node in the decision tree. The model shows limited sensitivity to the specified feature differences, suggesting these particular combinations map to similar historical patterns.

This finding has important implications. First, predictions should be interpreted as coarse approximations not precise forecasts. Second, the model is best used for directional guidance and scenario comparison rather than exact budgeting. Third, identical outputs limit practical utility for detailed decision making. Fourth, the model may need additional features such as marketing spend, competition, or franchise status to better differentiate. Fifth, more complex ensemble methods like random forests or gradient boosting might provide better granularity in future work.

	Movie	Predicted_Profit
0	Gorga	44,266,022.70
1	King Wolf	44,266,022.70
2	Hellish Gray	44,266,022.70

Predictions Table

Prediction Details:

- **Gorga** (R-rated Drama, $130M budget, June 2026 release): Predicted profit = $44.27 million
- **King Wolf** (R-rated Drama, $250M budget, March 2027 release): Predicted profit = $44.27 million
- **Hellish Gray** (PG-rated Comedy, $310M budget, July 2026 release): Predicted profit = $44.27 million

All three movies received identical predictions because they were classified into the same terminal node of the decision tree based on their feature values. Since we filled missing score and votes with dataset averages (score ≈ 6.33, votes ≈ 81,768), and the other features did not create enough differentiation in the tree structure, our model predicted the same outcome. This limitation suggests that while the model provides reasonable average predictions, it

lacks the granularity needed to distinguish between different types of films in this particular feature space.

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6. Managerial Implications and Recommendations

Based on our combined descriptive, regression, and predictive analyses, several actionable recommendations emerge for industry decision makers.

First, prioritize audience engagement over critical acclaim. Our finding that votes is the strongest profit predictor while score has much smaller effect suggests marketing should focus on reach, awareness, and social media engagement rather than awards campaigns when profit is the primary goal. Invest in grassroots marketing, influencer partnerships, and viral strategies. Track pre-release social metrics as leading indicators of commercial potential.

Second, favor Comedy and Horror genres for profitability. Only these two genres show significant positive profit effects in our regression, both around 20 million dollars advantage over baseline. When creative flexibility exists, develop properties in these genres as they offer favorable risk-return profiles. Animation shows strong descriptive profit but is not significant after controls, possibly because effects are confounded with budget. Avoid Family and Western genres unless specific strategic reasons exist.

Third, avoid fall release windows particularly September, October, and November. These months show significant profit penalties of 12 to 15 million dollars after controls. Reserve these months for niche releases or counter-programming. Move major tentpole releases to summer or holiday periods. Fall releases should only occur with strong competitive positioning or awards strategy considerations.

Fourth, focus on recent contemporary content. Each additional year since production reduces profit by approximately 1.2 million dollars. Prioritize timely stories, avoid long development-to-release cycles, fast-track promising projects to market, and update or refresh older properties before greenlight.

Fifth, control budgets carefully. Budget shows negative correlation with profit in our descriptive analysis. There is no evidence that higher spending guarantees proportional returns. Lean toward modest budgets unless clear tentpole justification exists. Every additional budget dollar must be justified by revenue model. Horror and Comedy succeed with moderate budgets, so avoid bloated production costs requiring unrealistic box office to recoup.

Sixth, ratings strategy is secondary to other factors. MPAA rating shows no significant profit effect after controls in our regression. Choose rating based on creative needs not profit optimization. Do not force PG-13 if it compromises artistic vision. Rating effects seen descriptively are explained by genre and budget choices. Focus on genre and timing optimization over rating manipulation.

Seventh, use predictive models as screening tools not precise forecasts. Our decision tree achieves only 27% R-squared with 41 million dollar RMSE, and predictions lack granularity with identical outputs for diverse films. Use models for portfolio-level risk assessment and to

flag high-risk projects early in development. Compare scenarios directionally such as March versus July release timing. Do not rely on models for precise profit forecasts. Supplement quantitative analysis with domain expertise and market intelligence.

Eighth, invest in better data collection and advanced methods. Our current models explain only 25 to 27% of variance. The neural network failed to converge properly. Collect data on marketing spend, competitive environment, and franchise status. Test ensemble methods like random forests and gradient boosting. Incorporate pre-release buzz metrics from social media and trailer views. Build separate models for domestic versus international markets. Consider time-series approaches for evolving industry dynamics.

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7. Limitations and Future Research

This analysis has several important limitations that should be acknowledged. The dataset is missing key variables likely to be very important drivers of profitability. Marketing and promotional budgets are probably among the strongest profit determinants but were not available. Distribution intensity measured by screen count and territories is absent. Competitive release environment is not quantified. Franchise and sequel indicators are not coded. Star power metrics based on prior box office performance are not included. Pre-release buzz from trailer views and social sentiment is unavailable.

Additionally, score and votes are post-release metrics rather than pre-release predictors, limiting their usefulness for pure forecasting applications. Real prediction would require pre-release proxies such as social media activity before opening weekend.

The sample is limited to IMDb-tracked films which may create selection bias toward higher-budget releases. Many independent and international films are missing. The profit measure based on theatrical gross does not capture revenues from home video, streaming, international markets beyond included gross figures, or merchandising which can be substantial.

The neural network showed severe underperformance requiring acknowledgment. The current configuration with 500 maximum iterations clearly failed to converge properly. Extensive hyperparameter tuning would be needed including learning rate, hidden layer architecture, activation functions, and regularization approaches. High-dimensional sparse data from many dummy variables may be particularly challenging for standard multi-layer perceptrons. Alternative architectures should be explored.

The decision tree while better than neural network also showed limitations with identical predictions for diverse films. Deeper trees or alternative split criteria might improve differentiation but risk overfitting. The fundamental challenge is that 27% explained variance leaves approximately three quarters unexplained, reflecting inherent unpredictability in creative industries.

Temporal considerations are also important. The model was trained on historical data up to around 2020. Structural industry shifts have occurred including rise of streaming platforms, shortened or eliminated theatrical windows especially post-pandemic, and changing consumer behavior. Historical patterns may not hold going forward, limiting model applicability to current environment.

Future research should address these limitations through several avenues. Enhanced modeling approaches should test gradient boosting algorithms like XGBoost and LightGBM, random forests for ensemble predictions, and properly tuned deep learning architectures. Causal inference methods such as instrumental variables and difference-in-differences could provide stronger effect isolation beyond correlational findings.

Additional features should be collected including marketing spend which is the strongest omitted variable, social media metrics covering mentions, sentiment, and engagement, franchise indicators and intellectual property valuation, star power indices, and competition intensity measures.

Refined analysis could develop separate models for domestic versus international markets, genre-specific models, time-varying coefficients to capture industry evolution, and explicit interaction effects between timing and genre. Validation approaches should include out-of-time testing where models trained on old data are tested on recent films, industry partnerships to access ground-truth marketing data, and potential A/B testing of release strategies.

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8. Conclusion

This project demonstrates that movie profitability while highly uncertain is systematically related to observable production and release characteristics. Our analysis provides both explanatory insights and predictive tools useful for decision making in the film industry.

The strongest drivers of profitability identified through our regression analysis are audience engagement measured by votes which dominates all other predictors, recency with newer releases significantly outperforming older films controlling for inflation, Comedy and Horror genres which show clear profit advantages after controls, and avoidance of fall months September through November which impose significant penalties. Many factors appearing important in descriptive analysis lose significance in multivariate regression.

Several factors show limited or no independent effects after controlling for confounds. MPAA rating has no significant effect after controls despite appearing to matter descriptively. Most genres show no significant differences in multivariate models with only 2 of 14 genres remaining significant. Summer release advantages seen descriptively are partly explained by types of films released rather than timing per se. Runtime and title length have minimal economic impact despite statistical significance.

Our predictive modeling exercise using machine learning approaches yielded mixed results. The decision tree achieved moderate success with 27% R-squared out-of-sample, slightly outperforming linear regression. However, the neural network completely failed to learn effectively with current configuration, achieving only 1% R-squared. All models leave substantial unexplained variance reflecting the fundamental unpredictability of entertainment.

The practical value of our analysis lies in providing both explanatory insight and predictive tools for portfolio structuring and project prioritization, genre and timing strategy formulation, budget discipline and resource allocation, and early-stage risk assessment. However, models should be used as decision support tools providing directional guidance

rather than definitive forecasts. Success in the film industry requires balancing data-driven insights with creative vision and market intuition.

Our final recommendation is to focus on maximizing audience engagement through marketing and word-of-mouth strategies, select financially favorable genres particularly Comedy and Horror when creative flexibility exists, avoid fall release windows for profit-seeking releases, control budgets carefully without assuming higher spending guarantees returns, and use predictive models for screening and scenario comparison rather than precise forecasting. Producers and investors should combine quantitative analysis with domain expertise, market intelligence, and recognition of the creative industries' inherent unpredictability to make informed decisions while managing the substantial risks that will always exist in film production.